Development Policymaking and the Roles of Market, State, and Civil Society

Chapter 11

Econ 322 – The Global Economy: Trade and Development

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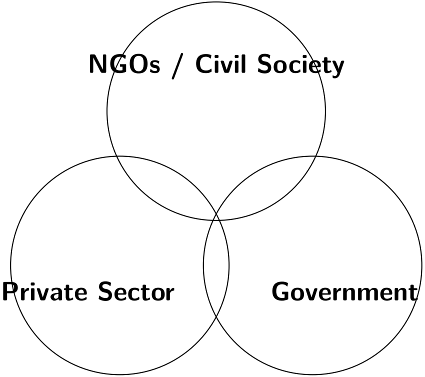
August 24, 2026

Based on Todaro, P. M. and Smith, C. S. (2020), *Economic Development*, 13th ed., Pearson.

11.1 A Question of Balance

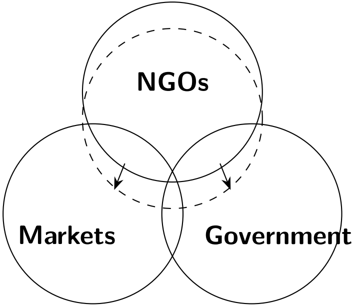
- Roles and limitations of state, market, and civil society/citizen sector/NGOs in achieving economic development and poverty reduction
- Venn diagrams:
  - Overlapping activities; and
  - Extensions of sector activity, illustrated with NGOs substituting for missing government or private sector activity

Venn Diagram: Sector Spheres and Overlaps



A. Sectoral overlap: Differentiation and zones of uncertainty on organizational comparative advantage

Venn Diagram – Sector Overlapping and Sector Extension: Example of NGOs



B. Sectoral extension: Extension beyond standard comparative advantage depending on specific circumstances

Source: Brinkerhoff, Smith and Teegen, "Beyond the 'Non': The Strategic Space for NGOs in Development," in NGOs and the Millennium Development Goals, Palgrave Macmillan, 2007, pp. 53–80 (Figure 11.3 in Todaro–Smith).

11.2 Development Planning: Concepts and Rationale

- The Planning Mystique
  - In the past, few doubted the importance and usefulness of national economic plans
  - Recently, however, disillusionment has set in
  - But a comprehensive development policy framework can play an important role in accelerating growth and reducing poverty

Why did the "planning mystique" collapse? Many post-independence plans were overambitious, ran on weak institutions and unreliable data, leaked resources through corruption, and could not adjust to shocks. Planning is not dead—it should *guide* markets, not replace them.

11.2 Development Planning: Concepts and Rationale (Continued)

- The nature of development planning resource mobilization for public investment
  - Economic policy to control private economic activity according to social objectives formulated by government
- Planning in mixed developing economies
- Private sector in mixed economies comprises
  - The subsistence sector
  - Small-scale businesses
  - Medium size enterprises
  - Larger domestic firms
  - Large joint or foreign owned enterprises

11.2 The Rationale for Development Planning

- Four classic arguments for a national plan:
  - Market failure
  - Resource mobilization and allocation
  - Attitudinal or psychological impact
  - Requirement to receive foreign aid

What each argument solves. *Market failure*: markets undersupply public goods and ignore the poor and long-run goals. *Resource mobilization*: governments can pool scarce capital and direct it to priorities. *Psychological impact*: a plan signals direction and coordinates expectations. *Foreign aid*: donors and the World Bank/IMF often require a strategic framework.

Three General Forms of Market Failure

- The market cannot function properly or no market exists
- The market exists but implies inefficient resource allocation
- More expansively: the market produces undesirable results as measured by social objectives other than the allocation of resources
  - Often items such as more equal income distribution, and "merit goods" such as health, are treated as separate rationales for policy, outside of economic efficiency

Market Failure: A Typology

- **Market failures** can occur when **social costs or benefits differ from private costs** or benefits of firms or consumers
- **Market power** (monopolistic, monopsonistic)
- **Public goods:** free riders cannot be excluded except possibly at high cost
- **Common property resources** with a lack of (informal or formal) regulation
- **Externalities:** agents do not have to pay all the costs of their activities, or are unable to receive all the benefits
- **Prisoners' Dilemmas** occur when agents are better off if others cooperate, but individual agents are better off "defecting."
- **Coordination failures** can occur when coordination is costly, e.g., where a Big Push may be needed
- Capital markets are particularly prone to failure

Market and Government Failure: Broader Arguments

- Government failure: in many cases, politicians and bureaucrats can be considered utility maximizers (for themselves), not public interest maximizers
- So we cannot jump to a conclusion that if economic theory says policy can fix market failures that it will do so in practice
- Analysis of incentives for government failure guides reform, e.g. civil service reform, constitution design
- Developing countries tend to have both high market failure and high government failures

11.3 The Development Planning Process: Some Basic Models

- Characteristics of the planning process
- Planning in stages: basic models
  - Aggregate growth models
  - Multisector input-output, social accounting, and CGE models
- Three stages of planning
  - Aggregate
  - Sectoral
  - Project

11.3 Aggregate Growth Models (1/5)

Aggregate Growth Models: Projecting Macro Variables

$$K(t) = c\,Y(t) \tag{11.1}$$

where

- $K(t)$ is capital stock at time t
- $Y(t)$ is output at time t
- c is the average and marginal **capital–output ratio**

Equivalently, $Y(t) = \frac{1}{c} K(t)$.

11.3 Aggregate Growth Models (2/5)

$$I(t) = K(t+1) - K(t) + \delta K(t) = sY = S(t) \tag{11.2}$$

where

- $I(t)$ is gross investment at time t
- s is the savings rate
- S is national savings
- δ is the depreciation rate

If g is the targeted rate of output growth, then

$$g = \frac{Y(t+1) - Y(t)}{Y(t)} = \frac{\Delta Y(t)}{Y(t)} \tag{11.3}$$

11.3 Aggregate Growth Models (3/5)

$$\frac{\Delta K}{K} = \frac{c\,\Delta K}{K} = \frac{(K/Y)\,\Delta Y}{K} = \frac{\Delta Y}{Y} \tag{11.4}$$

$$g = \frac{sY - \delta K}{K} = \frac{s}{c} - \delta \tag{11.5}$$

$$n + p = \frac{s}{c} - \delta \tag{11.6}$$

where n is the labor force growth rate and p is the growth rate of labor productivity.

Reading (11.5). Planners use $g = s/c - \delta$ backwards: pick a target growth rate g, know the capital–output ratio c and depreciation δ, and the equation tells you the saving rate s the economy must reach. If saving cannot rise, the alternative is to lower c (use capital more efficiently) or raise productivity growth p.

11.3 Aggregate Growth Models (4/5)

$$W + \pi = Y \tag{11.7}$$

where W and π are wage and profit incomes.

$$s_\pi \pi + s_W W = I \tag{11.8}$$

where s_π and s_W are the marginal propensities to save from profit and wage income.

Combining the pieces:

$$\begin{aligned} Y &= W + \pi \\ S &= s_W W + s_\pi \pi \\ S &= I \\ I &= c(g + \delta)\, Y \end{aligned}$$

$$\Rightarrow \quad s_W W + s_\pi \pi = c(g + \delta)\, Y$$

11.3 Aggregate Growth Models (5/5)

Substituting $W = Y - \pi$:

$$\begin{aligned} s_W(Y - \pi) + s_\pi \pi &= c(g + \delta)\, Y \\ s_W Y - s_W \pi + s_\pi \pi &= c(g + \delta)\, Y \\ s_W Y + (s_\pi - s_W)\, \pi &= c(g + \delta)\, Y \end{aligned}$$

Dividing by Y:

$$s_W + (s_\pi - s_W)\left(\frac{\pi}{Y}\right) = c(g + \delta)$$

$$c(g + \delta) = (s_\pi - s_W)\left(\frac{\pi}{Y}\right) + s_W$$

11.3 Multisector Models and Sectoral Projections

- Interindustry or input-output models
- Can be extended in 2 ways
  - Social accounting matrix (SAM) models where data from national accounts, balance of payments (BOP), and flow-of-funds databases is supplemented with household survey data
  - Computable general equilibrium (CGE) models where utility and production functions are estimated and impacts of policies are simulated

Multisector Models

- Input-Output models: **formal models** that divide the economy into sectors and trace the flow of interindustry **purchases (inputs)** and **sales (output).**
- A set of **simultaneous algebraic equations** expresses the **specific production processes** or technologies of each industry.
- For example: agriculture is a producer of output and uses inputs from other the manufacturing sector.
- These models track direct and indirect effects of changes in the demand for the product from an industry on the output, employment, and imports of all other industries.

Multisector Models and SAM

- Input-output matrices can be extended with data on factor payments, sources of household income, and household consumption.
- This is called a **Social Accounting Matrix**.
- Adds data from the system of national accounts, balance of payments, and flow-of-funds databases.
- Supplemented with household survey data.

SAM example

	Account	Commodities	Factors	Households	Government	Rest World	of Taxes	Savings-Investment	Total
Account		975							975
Commodities	499			332	43	163		121	1,157
Factors	458					3			460
Households			398		15	32			445
Government			39			2	61		101
Rest of World		156	23	1	0				181
Taxes	19	25		17					61
Savings-Investment				95	43	-18			121
Total	975	1,157	460	445	101	181	61	121	

CGE models

- **General Equilibrium Models** are an approach to SAM where economists create a system of equations with individuals and firms (in each sector).
- Individuals are assumed to **maximize utility or demand**.
- Production functions are assumed or estimated from national accounts.
- **GEM fits the SAM data** and allows simulating the (deterministic) impact of some factors on the economy.
- Martin Cicowiez and Marco Sanchez from the UN have slides with an example on this website.

Project Appraisal

- Project appraisal is **the quantitative analysis** of the profitability of investing **public or private funds in alternative projects**.
- **Cost-Benefit Analysis**: The actual and potential private and social costs of economic decisions are weighed against the actual and potential private and social benefits.
- Basic idea: To decide on the worth of projects involving public expenditure, it's necessary to **weigh the advantages (benefits) and the disadvantages (costs) to society as a whole**.
- It is common in the US beyond development economics.

Methodology

- Actual receipts are not a true measure of social benefits.
- Actual expenditure is not a true measure of social cost.
- Why?
- **Opportunity costs.** What would that investment make in a different project?
- Externalities: costs and benefits that diverge from the private decision making.
- Social Profits: The difference between social benefits and social costs.

Estimating Social Profits

- Specify the **objective function** to be maximized with some measurement (per capita consumption, income distribution, etc.)
- **Social measures of the unit values of all products**. These are called **accounting prices or shadow prices** of inputs and outputs and are different from market prices.
- Choose **a decision criterion** to reduce the stream of projected benefits and costs to an index.
- Shadow prices: prices that reflect the true opportunity cost of resources.
- Market prices: prices established by demand and supply.

Step 1: Setting objectives

- National Cohesion, Self-Reliance, Political Stability, Modernization, and Quality of Life do not have clear quantitative measurements.
- Economic Planners measure the social worth of a project in terms of their contribution to (present and future) **consumption**.
- Recently, income distribution has become an important objective.

Step 2. Computing shadow prices

- Why are market prices not a reflection of social benefits and costs?
- Inflation and currency overvaluation:
  - Price controls alter the relative price of goods and services.
  - Chronic inflation.
- Wage rates, capital costs, and unemployment.
  - Factor price distortions lead to underestimating costs.
- Tariffs, quotas, subsidies, and import substitution.
  - Distorts real wages and relative prices.
  - Generates rent seeking and distortions across sectors.
- Savings deficiency.
  - Governments should use lower discount rates because investment is suboptimal.
- The social rate of discount.
  - They are different from market interest rates. Governments optimizing future citizens should have lower rates of discount.

Step 3: Choosing Projects

- **Net present value (NPV)**: discount the stream of social benefits B_t and costs C_t at the social rate of discount r:

$$\text{NPV} = \sum_t \frac{B_t - C_t}{(1+r)^t}$$

- Decision rule: accept if $\text{NPV} > 0$; reject if $\text{NPV} < 0$
- **Internal rate of return (IRR)**: the discount rate at which $\text{NPV} = 0$; accept if the IRR exceeds the (social) rate of discount

Example. A project costs \$500 today and returns \$300 in each of the next two years. At $r = 10\%$:

$$\text{NPV} = -500 + \tfrac{300}{1.1} + \tfrac{300}{1.1^2} \approx +21$$

Positive, so approve. A project with low *private* returns may still have a high *social* IRR if it improves equity, health, or the environment.

Some notes and criticism

- The results of **development planning have been disappointing** – what went wrong?
- Government policy often tends to **increase rather than reduce the divergences between private and social valuations**
- **Deficiencies in the plans and implementation**
  - Grandiose in design, vague on specific policies for achieving stated objectives.
  - Overreliance on strong assumptions.
  - Failure to account for the heterogeneities of individual behavior. ("Unintended Consequences").

Some notes and criticism: planning by computer

- The Ninth Five-Year Plan of the Soviet Union was a set of economic goals designed to strengthen the national economy in the early 1970s.
- The USSR had 7,000 "super" computers in 1971.
- The goal? Industrial automation, econometrics, and national central planning.

Elbrus Soviet supercomputer, Polytechnical Museum, Moscow.

Some notes and criticism

- As late as 1989, the copy of Samuelson's Economics: An Introductory Analysis, stated:
- "The Soviet economy is proof that, contrary to what many skeptics had earlier believed, a socialist command economy can function and even thrive."

Source: P. Samuelson (1968)

Some notes and criticism

- "The curious task of economics is to demonstrate to men how little they really know about what they imagine they can design." F. Hayek
- Some of these approaches are rather antiquated and inspired discussions on national economic planning in the 1970s-1980s.
- The **extended use of CGE models and mathematical economic departments** in government aimed at estimating and setting prices.
- In the 1970s, there was a strong belief that **computers would allow for planning every sector of an economy**.

Friedrich A. Hayek (1899–1992), Nobel laureate 1974.

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Some notes and criticism

- Aggregate models and dynamic general equilibrium models (DSGE) continue to be useful to study price shocks or responses to policies.
- They are mathematically complex but not part of a planning strategy to set prices and allocate resources.
- Cost-benefit analysis continues to be useful but relies on strong assumptions.
- Most social projects and investments today plan empirical evaluations.
- (Empirical) Impact Evaluation studies based on Randomized Controlled Trials (RCTs) are popular across developing nations, and many countries have their departments.

11.4 Government Failure and Preferences for Markets over Planning

- The 1980s policy shift toward free markets
  - Deregulate industries, privatize state-owned firms, cut subsidies and price controls, open up to foreign trade and investment

Why the shift? After decades of planning, many countries still faced budget deficits, corruption and bureaucracy, shortages, and low growth despite heavy public investment. The diagnosis moved from "market failure" to *government failure*—but markets need strong institutions to function.

11.4 Problems of Plan Implementation and Plan Failure

- **Theory** versus **practice**
- **Deficiencies in the plans** and their implementation
- Insufficient and or **unreliable data**
- **Unanticipated economic disturbances**, external and internal
- **Institutional weaknesses**
- Lack of political will
- Conflict, post-conflict, and fragile states

11.5 The Market Economy

- Well functioning market economy requires
  - Clear property rights
  - Laws and courts
  - Freedom to establish business
  - Stable currency
  - Public supervision of natural monopolies
  - Provision of adequate information
  - Autonomous tastes
  - Public management of externalities
  - Stable monetary and fiscal policy instruments
  - Safety nets
  - Encouragement of innovation
  - Security from violence

11.5 The "Washington Consensus" and Its Limitations

- The "Washington Consensus" on the role of the state in development
- The consensus reflected a free market approach to development espoused by the IMF, the World Bank, and key U.S. government agencies
- However, it did not always accord with realities in developing countries that were successful

IMF headquarters, Washington, DC—a few blocks from the World Bank and the U.S. Treasury.

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Box 11.1 The Washington Consensus and East Asia

Elements of the Washington Consensus	South Korea	Taiwan
1. Fiscal discipline	Yes, generally	Yes
2. Redirection of public expenditure priorities toward health, education, and infrastructure	Yes	Yes
3. Tax reform, including the broadening of the tax base and cutting marginal tax rates	Yes, generally	Yes
4. Unified and competitive exchange rates	Yes (except for limited time periods)	Yes
5. Secure property rights	President Park started his rule in 1961 by imprisoning leading businessmen and threatening confiscation of their assets	Yes
6. Deregulation	Limited	Limited
7. Trade liberalisation	Limited until the 1980s	Limited until the 1980s
8. Privatisation	No: government established many public enterprises during 1950s and 1960s	No: government established many public enterprises during 1950s and 1960s
9. Elimination of barriers to direct foreign investment (DFI)	DFI heavily restricted	DFI subject to government control
10. Financial liberalisation	Limited until the 1980s	Limited until the 1980s

Source: Rodrik, Dani (1996), 'Understanding economic policy reform,' Journal of Economic Literature, 34: 17. Reprinted with permission from the American Economic Association and courtesy of Dani Rodrik.

11.6 The Washington Consensus and Its Subsequent Evolution

- Toward a new consensus
  - New emphasis on government's responsibility toward poverty alleviation and inclusive growth
  - Provision of fundamental public goods
  - Importance of health and education
  - A recognition that markets can fail
  - Governments can help secure conditions for an effective market-based economy

Hyundai Heavy Industries shipyard, Ulsan. South Korea's industrialization was state-guided and diverged from the Consensus (Box 11.1).

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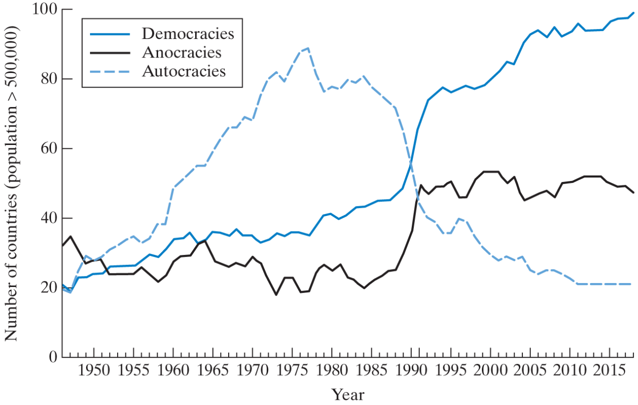
11.7 Development Political Economy: Theories of Policy Formulation and Reform

- Understanding voting patterns on policy reform
- Institutions and path dependency
- Democracy versus autocracy: which facilitates faster growth?

Path dependency: where an economy is today is shaped by decisions and institutions inherited from the past (e.g., colonial tax systems, entrenched landowning elites). Institutions do not simply emerge—they evolve slowly and politically, so reforms must be politically viable, not just economically efficient.

Figure 11.1 Global Trends in Governance, 1947–2017



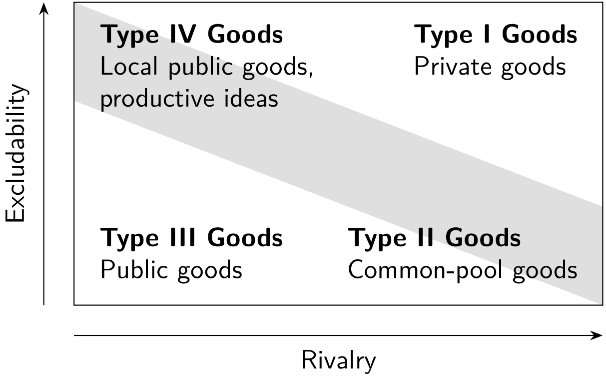
Note: The figure shows the percent of countries under each regime type. An anocracy is a mixed, or incoherent, authority regime.

Source: Monty G. Marshall and Benjamin R. Cole, Center for Systemic Peace, 2019. Reprinted with permission from the Center for Systemic Peace. See http://www.systemicpeace.org/vlibrary/GlobalReport2017.pdf

11.8 Development Roles of NGOs and the Broader Citizen Sector

- Also referred to as civil society, or the "voluntary sector"
- Potentially important roles in:
  - Common property resource management
  - Local (such as village or neighborhood level) public goods
  - Economic and productive ideas
- These activities are either:
  - Excludable but not rival
  - Rival but not excludable
  - Partly excludable and partly rival

Figure 11.2 Typology of Goods



The shaded diagonal indicates the area of primary NGO comparative advantage in dimensions of rivalry and excludability. When, based on local conditions (such as government failure), NGOs are in a position to supply public or private goods at a lower price or higher quality, they may be found expanding into the nonshaded areas as well (Type I and Type III goods).

11.8 Development Roles of NGOs and the Broader Citizen Sector (Continued)

- Rivalry and excludability are two characteristics of services; but not the only relevant ones
- Other potential comparative advantages of NGOs
  - Innovative design and implementation
  - Program flexibility
  - Specialized technical knowledge
  - Provision of targeted local public goods
  - Common-property resource management design and implementation
  - Trust and Credibility
  - Representation and advocacy
- But NGOs are subject to voluntary sector “failures”—just as there is market failure and government failure

Other limitations: “voluntary failure” – NGOs may be...

- Called voluntary failures, referring to the term “voluntary sector”
- Insignificant, owing to small scale and reach.
- Lacking necessary local knowledge to develop and implement an appropriate mix of programs to address relevant problems
- Selective and exclusionary, elitist, and or ineffective
- Lacking adequate incentives to ensure effectiveness
- Captured by goals of funders rather than intended beneficiaries; may change priorities one year to the next
- Giving too little attention to means, preventing needed scale...
- Or, means - such as fundraising - can become ends in themselves
- Lacking direct feedback (as private firms get in markets, or elected governments receive at the polls); this can let the weaknesses go on for some time before being corrected

11.9 Trends in Governance and Reform

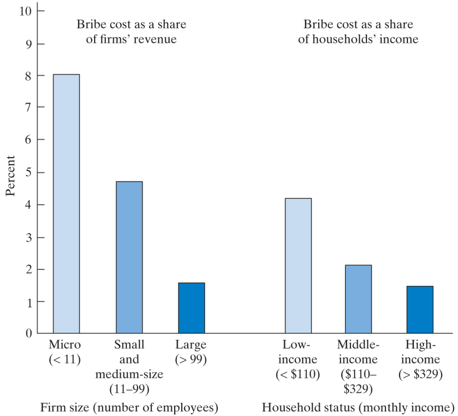
- Tackling the problem of Corruption
- Decentralization
- Development participation—alternate interpretations
  - Genuine participation and role of NGOs

Token vs. genuine participation. *Token*: people attend a workshop or fill out a survey, but decisions are already made. *Genuine*: communities help design, implement, and monitor projects. Decentralization brings decisions closer to the people, but local elites may capture power and local capacity may be weak.

11.9 Trends in Governance and Reform (Continued)

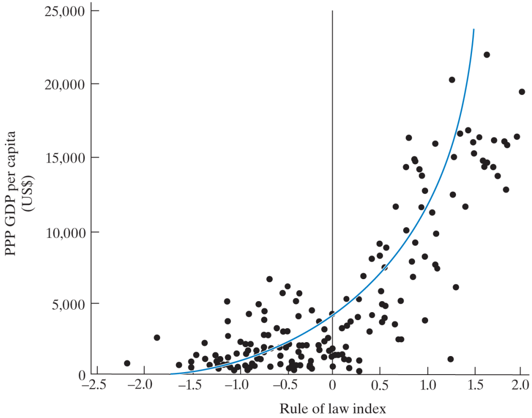
- Tackling the problem of corruption
  - Corruption is the abuse of public trust for private gain
  - A form of theft - even if indirectly
- Corruption has efficiency costs; even if it is less often the "binding constraint" on growth than often guessed
- Effects of corruption fall disproportionately on the poor
- Good governance is broader than simply an absence of corruption

Figure 11.4 Corruption as a Regressive Tax: The Case of Ecuador



Source: World Development Report, 2000–2001: Attacking Poverty, by World Bank, p. 102, fig. 6.2. Copyright © 2000 by World Bank. Reproduced with permission.

Figure 11.5 The Association Between Rule of Law and Per Capita Income



Source: World Development Report, 2000–2001: Attacking Poverty, by World Bank, p. 103, fig. 6.3. Copyright © 2000 by World Bank. Reproduced with permission.

Chapter summary

- State, market, and civil society each have a role; their activities overlap, and NGOs often extend into gaps left by government or market failure
- Planning is justified by market failure, resource mobilization, psychological impact, and aid requirements—but government failure is just as real in developing countries
- Planning tools: aggregate growth models ($g = s/c - \delta$), input-output/SAM/CGE models, and project appraisal using shadow prices, NPV, and IRR
- The 1980s Washington Consensus favored markets; East Asia's success only partly followed it, and a new consensus stresses public goods, health, education, and inclusive growth
- NGOs have comparative advantages (trust, flexibility, local public goods) but face "voluntary failure"; governance reform targets corruption, decentralization, and genuine participation

Concepts for Review

- Accounting prices
- Aggregate growth model
- Comprehensive plan
- Corruption
- Cost-benefit analysis
- Economic infrastructure
- Economic plan
- Economic planning
- Government failure
- Input-output model
- Interindustry model
- Internal rate of return
- Market failure

Concepts for Review (Continued)

- Market prices
- Net present value
- Nongovernmental organization (NGO)
- Partial plan
- Path dependency
- Planning process
- Political will
- Project appraisal
- Rent seeking
- Shadow prices
- Social profit
- Social rate of discount
- Voluntary failure